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Adversarial Training and Gradient Optimization for Partially Deepfake Audio Localization

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***Abstract*—Partially deepfake audio localization is important in audio forensics. However, existing localization models for partially deepfake audio face two major challenges: distribution shifts between training and testing data as well as insufficient utilization of information from both manipulated regions and boundaries. To address these challenges, we propose to use Adversarial training and Gradient Optimization (AGO) to improve partially fake audio localization. Specifically, we apply a gradient reversal layer to reduce the dependence on domain-specific features, enhancing the model's generalization ability. Additionally, we introduce an alternating update strategy to learn information from both manipulated regions and boundaries, while orthogonal gradient updates minimize conflicts between the two tasks. We evaluated AGO on both the ADD2023 track 2 and PartialSpoof datasets. We achieved a 22.82% relative improvement over the first-ranked method of the ADD2023 track 2. We also achieved state-of-the-art results on the PartialSpoof dataset. Our code is available at https://github.com/Little-dingding/ATGO.**

***Index Terms*—Partially deepfake audio localization, adversarial training.**

I. Introduction

Although advancements in speech synthesis and voice conversion have brought about convenience, they have also introduced the security risk of audio deepfake [1]–[3]. Partially deepfake audio, as a more covert and subtle type of audio deepfake, has recently attracted growing attention. In audio forensics, it is crucial to precisely identify and localize manipulated regions within partially deepfake audio, as these regions can provide vital information and evidence [4].

Existing partially deepfake audio localization models are primarily categorized into two tasks: manipulated region detection [5]–[7] and boundary detection [8]–[10]. As shown in Fig. 1, manipulated region detection focuses on identifying variations between manipulated and non-manipulated regions, while boundary detection leverages splicing traces left by manipulation to distinguish between boundary frames and non-boundary frames. Most current methods adopt one of the two tasks for PFA localization, either through boundary detection models or spoof detection models. Cai et al. [11] employed VAE [12] to fuse the results of both models for prediction. There are still areas for improvement. Since boundary information can assist in the localization of manipulated regions, while information from manipulated regions can help reduce false positives in boundary detection, it is necessary for the two types of information to mutually reinforce each other during the training process.

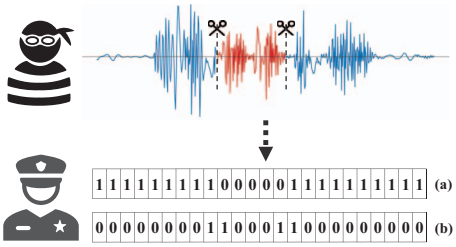

Fig. 1. The labeling method of partially deepfake audio for the manipulated region detection (a) and the boundaries detection (b).

Apart from improved localization, the domain shift commonly observed in partially deepfake audio across datasets also needs to be addressed [13]. Since manipulated segments and non-manipulated segments of partially deepfake audio generally come from different distributions, the model can easily learn to distinguish their differences using domain-specific features and perform well on datasets within the same domain. However, these domain-specific features can't represent the manipulated/non-manipulated segments of other distributions, causing the model to fail in real-world scenarios. To this end, previous works often leveraged standard data augmentation methods to enhance the generalization abilities of partially deepfake audio localization models, such as re-splicing audio, adding noise, and processing audio segments with new voice conversion models [11], [14], [15].

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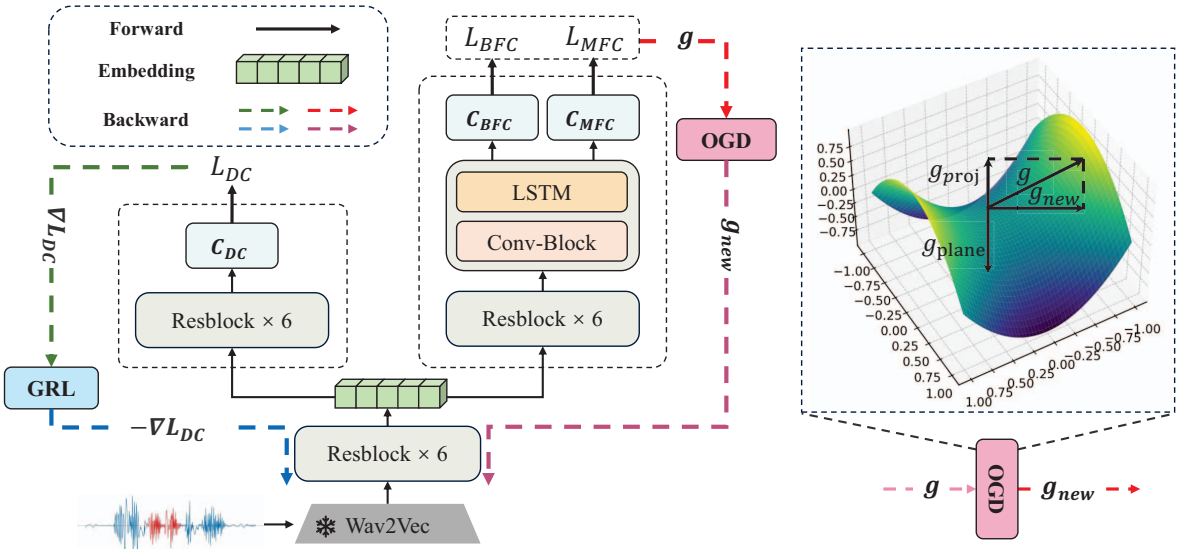

Fig. 2. Illustration of the AGO Architecture. (i) The encoder takes features from the Wav2Vec model, producing enhanced feature representations that are passed to a dual-branch network. C_{DC} stands for domain classifier, and for the domain classification loss L_{DC}, the gradient is reversed by the GRL during backpropagation, making the encoder reduce dependency on domain-specific features. (ii) For the manipulated region detection and boundary detection, each task has its own classifier: C_{BFC} (Boundary Frame Classifier) and C_{MFC} (Manipulated Frame Classifier). The losses for these tasks, L_{BFC} and L_{MFC}, are alternately backpropagated, and orthogonal gradient descent is used to adjust the gradient direction to minimize conflicts between the two tasks.

However, these standard methods still have limitations when dealing with datasets of specific distributions [16], [17], such as datasets of new types of attacks.

To address the domain shift and utilize the information from both manipulated regions and boundaries, we propose to use Adversarial training and Gradient Optimization (AGO) for partially deepfake audio localization. Adversarial training makes the model domain-invariant, and Gradient Optimization improves partially deepfake audio localization by reducing conflicts during the learning of both types of information. Specifically, we employ two branch networks: one is designed for domain classification, while the other handles two detection tasks—manipulated region detection and boundary detection. The domain classification branch uses a gradient reversal layer (GRL) to minimize its dependency on domain-specific features [18]. The branch of two detection tasks has two classifiers: one focuses on distinguishing between manipulated and non-manipulated regions, while the other specializes in detecting the boundaries of manipulated regions. We alternately update the two detection tasks, leveraging orthogonal gradient descent (OGD) [19] to minimize interference between their gradient updates, allowing for effective information sharing and joint optimization.

We evaluated AGO on two mainstream partially deepfake audio datasets, ADD2023 track 2 [20] and PartialSpoof [21]. The results show that AGO provides a 22.82% relative improvement over the first-ranked method of the ADD2023 track 2, we also achieved new state-of-the-art results on the PartialSpoof dataset over existing partially deepfake audio localization methods. Our main contributions are as follows:

- We propose a framework based on adversarial training for partially deepfake audio localization, which effectively enhances the localization capability of deep learning models under cross-domain conditions.
- Utilizing OGD, we designed an alternating update strategy to reduce conflicts between manipulated region detection and boundary detection, effectively integrating the information from both tasks for improved partially deepfake audio localization.

II. Proposed Method AGO

As illustrated in Fig. 2, our method (AGO) employs a dual-branch architecture where both branches share a common feature extractor. One branch is responsible for domain classification, while the other handles two detection tasks—manipulated region detection and boundary detection. We aim to train an encoder that is robust to domain variations, thereby reducing the model's reliance on domain-specific features. The encoder consists of six residual blocks and takes audio features extracted by Wav2Vec as input, enhancing the feature representations for downstream tasks.

A. Adversarial Training with GRL

For the domain classification, we utilize six ResBlocks as branch network B_{DC} and a fully connected layer as the domain classifier C_{DC}. The domain classification loss L_{DC}, based on Binary Cross-Entropy Loss (BCELoss) [22], is calculated as Eq. (1):

$$L_{\text{DC}} = \text{BCE}(C_{\text{DC}}(B_{\text{DC}}(F_{\text{feat}}(x_i))), d_i) \tag{1}$$

where d_i signifies the domain label for the i-th frame, and $F_{\text{feat}}(x_i)$ is the feature extracted from the input x_i.

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A GRL is employed to reverse the gradient from the domain classifier back to the encoder, fostering domain invariance during the training process. The gradient reversal can be expressed as Eq. (2):

$$\nabla L_{\text{DC}} = -\nabla L_{\text{DC}} \tag{2}$$

B. Gradient Optimization with OGD

Since boundary information can assist in the localization of manipulated regions, and information from manipulated regions can help reduce false positives in boundary detection, we train a backbone network capable of detecting both manipulated regions and boundaries. This enables joint predictions and improves localization accuracy.

The tasks of manipulated region detection and boundary detection share the same backbone, which includes the encoder $F_{\text{feat}}(x_i)$ and the branch network B_{L}. B_{L} consists of six ResBlocks, a convolutional block, and an LSTM [23] layer. Each task concludes with dedicated classifiers. The BCELoss of both tasks is calculated as Eq. (3) and Eq. (4):

$$L_{\text{MFC}} = \text{BCE}(C_{\text{MFC}}(B_{\text{L}}(F_{\text{feat}}(x_i))), y_i) \tag{3}$$

$$L_{\text{BFC}} = \text{BCE}(C_{\text{BFC}}(B_{\text{L}}(F_{\text{feat}}(x_i))), b_i) \tag{4}$$

where y_i represents the label of i-th frame for manipulated region detection, and b_i is the label for boundary detection. C_{MFC} is the classifier for the manipulated region detection, while C_{BFC} is the classifier for boundary detection.

We propose the OGD method to update the two tasks alternately.

For the input partially deepfake audio, labels are determined based on the manipulated regions. In the alternating update, we compute the gradient g for the manipulated region detection or boundary detection as Eq. (5):

$$g = \begin{cases} \nabla_\theta L_{\text{MFC}}(\theta, D) \\ \nabla_\theta L_{\text{BFC}}(\theta, D) \end{cases} \tag{5}$$

where g represents the gradient for the current task (manipulated region detection or boundary detection), ∇_θ represents the gradient with respect to the parameters θ of the backbone (the encoder $F_{\text{feat}}(x_i)$ and the branch network B_{L}), and D is the dataset for the current task.

After computing the gradient g for the current task, we project it onto the saved gradient plane g_{plane} (Eq. (6)), which represents the gradient plane updated from the previous task. The orthogonal gradient relative to the current gradient plane is calculated as Eq. (7):

$$g_{\text{proj}} = \frac{g \cdot g_{\text{plane}}}{\|g_{\text{plane}}\|^2} g_{\text{plane}} \tag{6}$$

$$g_{\text{new}} = g - g_{\text{proj}} \tag{7}$$

where g_{new} is the newly computed orthogonal gradient for the current task, and g_{plane} refers to the gradient plane from the previous task.

We then update the model parameters using the orthogonal gradient as Eq. (8):

$$\theta = \theta - \eta \cdot g_{\text{new}} \tag{8}$$

where η is the learning rate used for gradient descent. The updated orthogonal gradient g_{new} is then saved as the new gradient plane (Eq. (9)).

$$g_{\text{plane}} = g_{\text{new}} \tag{9}$$

This approach ensures that gradient updates between different tasks are as non-interfering as possible, thereby improving the ability to localize partially deepfake audio.

C. Total Loss Function

During the training process, the model accepts labeled data from the source domain and unlabeled data from the target domain. The source domain is used for the branch of manipulated region detection and boundary detection, while both the source and target domain data are passed through the branch for domain classification simultaneously. The total loss function L_{Total} is given by Eq. (10):

$$L_{\text{Total}} = L_{\text{MFC}} + L_{\text{BFC}} + \alpha \cdot L_{\text{DC}} \tag{10}$$

where α is a hyperparameter that controls the relative importance of the domain classification loss L_{DC}.

III. Experiment Setup

We evaluate our approach on two mainstream partially deepfake audio datasets: PartialSpoof [21] dataset and ADD2023 track 2 (ADD) dataset [20]. PartialSpoof has 25380 samples in the training set and 71237 samples in the test set. The ADD2023 challenge [20] introduced a partially deepfake audio Localization track, emphasizing the evaluation of partially deepfake audio localization models in cross-domain scenarios. The training set of ADD is derived from the HAD dataset [24], while the test set comprises 50,000 samples of genuine and partially deepfake audio from six diverse datasets, each of which includes added noise. For adversarial training, we randomly selected 1/10 of the unlabeled test samples from two datasets for access, reserving all samples for evaluation.

We employ mainstream evaluation metrics to assess model performance: Accuracy (Acc) and F_1 [22], and supplement with additional metrics such as EER, Precision, Recall, and Score [20] as used in various other papers.

We trained the model using the Adam optimizer with a learning rate η of 0.0001 and a batch size of 16 on an NVIDIA A100 80GB GPU. The parameters of Wav2vec 2.0 were initialized from the pre-trained model XLSR-53, and the minimal frame of input audio was set to 40ms. The parameter α of the total loss function is set to 0.5.

IV. Results and Analysis

A. Main Results

We selected the network architecture of RSDM [8] and SPF [25] for comparison, each of which currently performs best on the HAD and ADD datasets, respectively. The model architecture of SPF consists of ResBlocks and LSTM [23], while RSDM combines LCNN [26] with LSTM.

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TABLE I
THE $Acc(\%)$, $Precision(\%)$, $Recall(\%)$, $F_1(\%)$, AND $Score(\%)$ RESULTS OF THREE MODELS (RSDM, SPF, AND OURS) ON TWO DATASETS (PARTIALSPOOF AND ADD2023 TRACK 2).

	PartialSpoof					ADD2023 track 2				
Method	$Acc\uparrow$	$Precision\uparrow$	$Recall\uparrow$	$F_1\uparrow$	$Score\uparrow$	$Acc\uparrow$	$Precision\uparrow$	$Recall\uparrow$	$F_1\uparrow$	$Score\uparrow$
RSDM	76.98	72.15	63.87	67.76	78.43	79.07	75.32	62.65	68.40	78.67
SPF	86.34	76.80	72.47	74.57	83.63	74.56	52.14	**67.89**	58.98	66.28
AGO (ours)	**96.52**	**97.21**	**91.67**	**94.36**	**95.87**	**86.98**	**82.15**	63.87	**71.87**	**82.45**

TABLE II
THE $Acc(\%)$, $F_1(\%)$, AND $Score(\%)$ RESULTS OF OUR METHOD (AGO) ON THE ADD 2023 TRACK 2 DATASET OF THE ABLATION STUDY.

Model	$Acc\uparrow$	$F_1\uparrow$	$Score\uparrow$
AGO (ours)	**86.98**	**71.87**	**82.45**
w/o GO	81.34	65.36	76.55
w/o AT	79.52	57.03	72.77
w/o GO & AT	77.03	52.66	69.72
Boundary detection	57.49	46.18	54.10
Manipulated region detection	49.12	43.14	47.19

TABLE III
THE $Acc(\%)$, $F_1(\%)$, AND $Score(\%)$ RESULTS OF VARIOUS MODELS ON THE ADD 2023 TRACK 2 DATASET.

Team	$Acc\uparrow$	$F_1\uparrow$	$Score\uparrow$	**Data Aug.**
C01 [11]	82.23	60.66	67.13	Noise; reverb
C02 [5]	80.91	54.60	62.49	Noise; re-splicing
C03 [7]	79.56	54.50	62.42	Noise; reverb
C04 [27]	78.16	51.67	59.62	-
C05 [28]	74.52	52.53	59.12	Noise
AGO (ours)	**86.98**	**71.87**	**82.45**	-

TABLE IV
THE $EER(\%)$, $Precision(\%)$, $Recalll(\%)$, AND $F_1(\%)$ RESULTS OF VARIOUS MODELS ON THE PARTIALSPOOF DATASET.

Model	$EER\downarrow$	$Precision\uparrow$	$Recall\uparrow$	$F_1\uparrow$
IFBDN [29]	9.68	93.72	90.32	91.99
PSDL [30]	12.47	91.82	87.53	89.62
CFPRF [31]	7.41	95.23	**92.59**	93.89
AGO (ours)	**6.79**	**97.21**	91.67	**94.36**

Table I clearly demonstrates the advantages of our method across multiple evaluation metrics, particularly its generalization ability under cross-domain conditions. Our model achieved an accuracy of 96.52% and an F1 score of 94.36% on the PartialSpoof dataset, outperforming other methods. Our method also demonstrated strong performance on the ADD dataset, with an accuracy of 86.98% and an F1 score of 71.87%.

B. Ablation Study

As shown in Table II, we evaluated the contributions of adversarial training (AT) and gradient optimization (GO) to the model's performance. The results show that removing AT led to a drop in accuracy from 86.98% to 79.52% and a decline in F1 score from 71.87% to 57.03%, indicating that AT plays a crucial role in mitigating domain shifts and enhancing the model's generalization across different datasets. When GO was removed, the accuracy decreased to 81.34% and the F1 score dropped to 65.36%, demonstrating GO's effectiveness in alleviating gradient conflicts between the manipulated region detection and boundary detection tasks, thereby improving localization performance. The combined removal of AT and GO resulted in the most significant performance degradation, with accuracy falling to 77.03% and the F1 score to 52.66%. These findings highlight that the integration of AT and GO significantly enhances the model's robustness in cross-domain tasks and facilitates efficient information sharing between tasks. Furthermore, the single-task results underscore the significance of task collaboration.

C. Comparison with Other Results

We compared our results on the ADD 2023 track 2 dataset with those of the top five rankings in ADD 2023 track 2. As shown in Table III, our method achieved an F1 score of 71.87%, representing a substantial improvement with a relative increase of 22.82% over the first-ranked result, without using any data augmentation methods. We also compared our results on the PartialSpoof dataset with the best-performing methods currently available on PartialSpoof. As shown in Table IV, AGO records an EER of 6.79% and an F1 score of 94.36%, outperforming models like IFBDN and PSDL. These results demonstrate the effectiveness of our method, particularly showing significant improvements in the cross-domain ADD dataset.

V. CONCLUSION

We introduced AGO, a novel method for partially deepfake audio localization, which integrates adversarial training with a gradient reversal layer to enhance the model's generalization under cross-domain conditions. By employing gradient optimization with orthogonal gradient descent, AGO effectively mitigates conflicts between manipulated region detection and boundary detection, allowing for seamless fusion of critical information from both tasks to enhance localization accuracy. The experiment results of AGO demonstrate a 22.82% improvement over the first-ranked method on the ADD2023 Track 2 dataset and establish a new state-of-the-art on the PartialSpoof dataset, affirming the robustness and effectiveness of our approach.

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